Algebra: Operations with Complex Numbers

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N

Name: _____ Date: _____ Score: / 10

DIRECTIONS *Simplify powers of i. Add, subtract, or multiply complex numbers. Simplify fully.*

1 Simplify the power of i:

$$i^2$$

Answer: _____

2 Add the complex numbers:

$$(4 + 1i) + (3 - 4i)$$

Answer: _____

3 Solve:

$$x^2 + 9 = 0$$

Answer: _____

4 Add the complex numbers:

$$(-4 - 4i) + (-4 - 1i)$$

Answer: _____

5 Simplify the power of i:

$$i^7$$

Answer: _____

6 Multiply the complex numbers:

$$(0 - 1i)(2 - 1i)$$

Answer: _____

7 Add the complex numbers:

$$(-4 - 1i) + (0 + 1i)$$

Answer: _____

8 Solve:

$$x^2 + 25 = 0$$

Answer: _____

9 Multiply the complex numbers:

$$(-3 + 1i)(-2 - 3i)$$

Answer:

10 Simplify the power of i:

$$i^4$$

Answer:

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Answer Key & Solutions

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TEACHER NOTES *$i^2=-1$, $i^3=-i$, $i^4=1$, then cycles. For products: FOIL then replace $i^2=-1$.*

1 Simplify the power of i:

i^2

= -1

2 Add the complex numbers:

$(4+1i)+(3-4i)$

= $7-3i$

3 Solve:

$x^2+9=0$

= $x=\pm 3i$

4 Add the complex numbers:

$(-4-4i)+(-4-1i)$

= $-8-5i$

5 Simplify the power of i:

i^7

= $-i$

6 Multiply the complex numbers:

$(0-1i)(2-1i)$

= $-1-2i$

7 Add the complex numbers:

$(-4-1i)+(0+1i)$

= $-4+0i$

8 Solve:

$x^2+25=0$

= $x=\pm 5i$

9 Multiply the complex numbers:

$(-3+1i)(-2-3i)$

= $9+7i$

10 Simplify the power of i:

i^4

= 1

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